Simulation-Based Wrist Exoskeleton Assistance after Stroke: Optimal Control vs. Reinforcement Learning



Concept Presentation

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Introduction:

After a stroke, weak wrist extensors make the wrist drop. Everyday tasks such as lifting a cup and holding it level start to fail. A wrist exoskeleton can hold the wrist up, but how much torque does it need, and is a simple spring enough or is a motor required?

Key motivations:

- Stroke survivors rank arm and hand function among their top rehabilitation priorities
- Size wrist assistance in simulation before building hardware
- Compare an optimal plan with a learned, robust controller

Project Scope:

Aim:
Find the minimum wrist assistance a weakened, post-stroke wrist needs to lift and hold a cup, compare a spring with a motor, and test whether a learned controller keeps it working under real-world variation.

Includes:

- Predictive cup lift-and-hold simulation (OpenSim Moco, MoBL-ARMS model)
- Wrist-extensor weakness sweep; spring vs. motor assistance
- MyoSuite (MuJoCo) wrist-hand environment with PPO training
- Robustness tests and evaluation with defined metrics

Out of Scope (Current Phase):

- Real exoskeleton hardware and experiments with patients
- Spasticity modelling, finger rehabilitation and clinical validation

Objectives:

Research question: How weak can the wrist extensors get before a simulated arm can no longer lift and hold a cup, and what is the minimum wrist assistance that restores it?

- Reproduce a validated, unassisted cup lift-and-hold simulation as the baseline
- Find the extensor-strength threshold below which the task fails
- Determine the minimum wrist torque that restores the task
- Compare a passive spring with an active motor
- Route B: train an RL policy and compare its robustness with the Moco plan

Related Work and Gap:

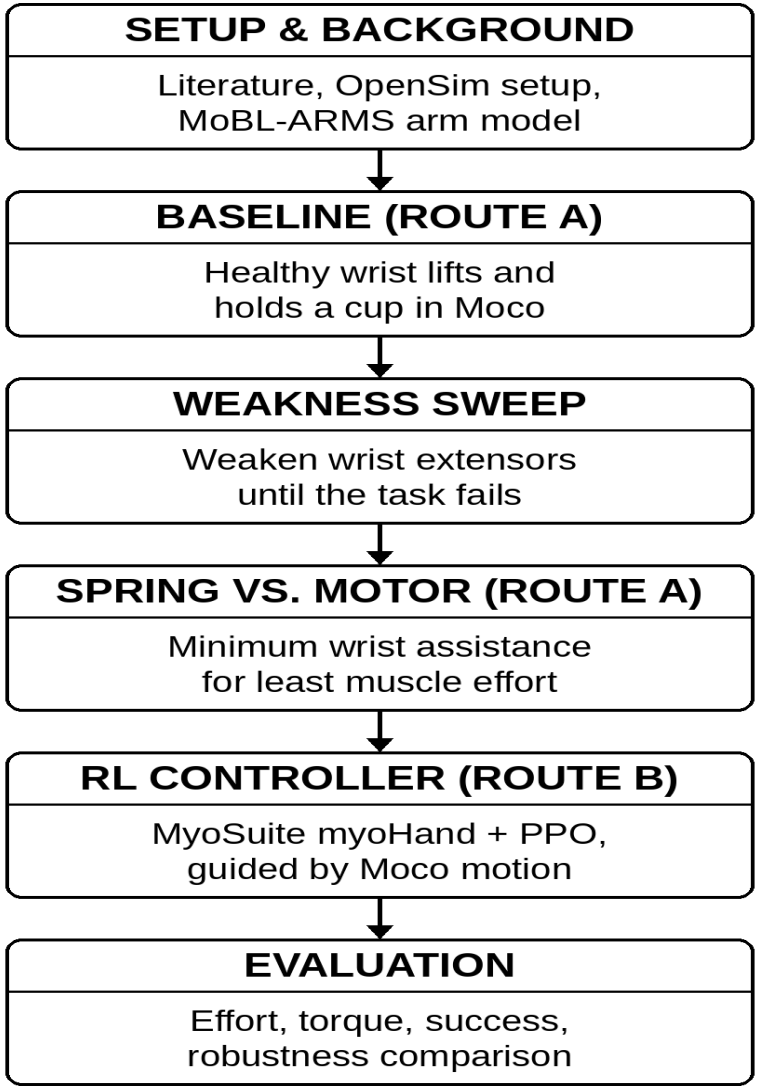
Work	What it did	Missing for our question
KhalilianMotamed Bonab et al., 2024 (Robotics)	OpenSim simulation of an elbow–wrist exosuit in a load-lifting task	Healthy subjects; no weakness, no learned control
Caggiano et al., 2022 (MyoSuite)	MuJoCo wrist/hand models with exoskeleton, sarcopenia and fatigue options	No wrist-assistance sizing study
Chavarrías et al., 2025 (arXiv)	RL exoskeleton torque controller trained in MyoSuite	Knee, not wrist

Gap: we found no study that sizes wrist assistance for a weakened, post-stroke wrist, or compares an optimal plan with a learned controller.

Technology Stack:

Process	Tool / Framework	Purpose
Musculoskeletal modelling	OpenSim 4.x, MoBL-ARMS upper-limb model	Arm and wrist model, weakness scaling
Optimal control (Route A)	OpenSim Moco, CasADi + IPOPT	Predict motion, optimise assistance
Physics simulation (Route B)	MuJoCo + MyoSuite (myoHand)	Wrist-hand cup task environment
Reinforcement learning	Stable-Baselines3 (PPO)	Train feedback controller
Evaluation & visualisation	Python: NumPy, pandas, Matplotlib	Metrics and plots
Version control	Git / GitHub	Reproducible runs

Project flow chart:



Working Methodology – Route A:

Step 1: Setup and Baseline
- Install OpenSim, load the MoBL-ARMS upper-limb model, reproduce a Moco example
- Simulate a healthy wrist lifting and holding a cup; check muscle patterns against literature

Step 2: Weakness Sweep
- Scale wrist-extensor strength (ECRL, ECRB, ECU) down in steps
- Find where the task fails (penalised reserve actuators separate this from solver failure)

Step 3: Spring vs. Motor
- Add a passive spring, then an ideal motor, at the wrist
- Minimise muscle effort; sweep spring stiffness and peak-torque limit

Step 4: Evaluation
- Muscle effort, peak extensor activation, task success, required torque
- All reported relative to the unassisted baseline

Working Methodology – Route B:

Step 1: Environment
- MyoSuite myoHand: cup lift-and-hold task with a wrist assist actuator
- Fingers held in a fixed grasp to keep the problem small

Step 2: Training
- PPO (Stable-Baselines3); reward tracks the Moco motion, rewards a level cup, penalises effort
- Policy outputs muscle and device commands together

Step 3: Full Hand Control (stretch goal)
- Release the fixed grasp and let the policy control the fingers too
- Only if Steps 1–2 are stable before the demo

Step 4: Robustness and Fallback
- Moco plan vs. RL policy with a heavier cup, pushes and MyoSuite muscle fatigue
- Fallback if training does not converge: reward-design analysis

Experimental Design:

Block	Setup	Purpose
A · Baseline	100% strength, no device	Validate against published patterns
B · Weakness sweep	Extensor strength reduced stepwise, no device	Threshold where the task fails
C · Assisted	Below threshold; spring or motor	Minimum assistance that restores the task
D · Sensitivity	Range of stiffness and peak-torque limits	Device torque vs. muscle effort
E · Robustness (Route B)	Moco plan vs. RL policy; cup mass, pushes, fatigue	Success under perturbations

Bill of Materials(BOM):

Component	Specifications	Quantity	Cost
GPU compute	THD GPU / cluster (to request)	Semester	Free
Cloud GPU (fallback)	Only if THD access unavailable	If needed	TBD
Models	MoBL-ARMS (OpenSim) and myoHand (MyoSuite)	-	Free
Software	Open source (OpenSim, MuJoCo, SB3)	-	Free
Data	No human-subject data needed	-	Free

Expected Outcomes:

- **K1:** extensor strength threshold (% of max. force) where lifting and holding the cup fails
- **K2:** minimum wrist assist torque (Nm) that restores the task, and whether a spring is enough
- **K3:** wrist-muscle effort reduction (%) compared with no assistance
- **K4:** RL policy success rate (%) vs. the Moco plan with a heavier cup, pushes and fatigue
- Reproducible pipelines for both routes, published on GitHub

Risks and Mitigation:

Risk	Route	Mitigation
Moco solver fails to converge	A	Lock shoulder and elbow; warm start from baseline
Solver failure mistaken for "task fails"	A	Penalised reserve actuators measure the deficit
RL policy never learns the task	B	Moco reference, fixed grasp, agreed fallback
OpenSim vs. MyoSuite model mismatch	B	Compare trends, not absolute values
Heavy compute for RL training	B	Request THD GPU / cluster early
Tight schedule: demo on 16 Dec	A + B	Route A first; full hand control only as stretch goal

Safety & Ethical Checklist:

- Simulation-only environment; no hardware, no human participants
- No personal data; open musculoskeletal models only
- Results are relative findings from a generic model, not clinical claims
- Model bias stated: one generic body type does not represent all users
- Reproducible and attributed: code, configs and seeds in Git; all sources cited

Future Work:

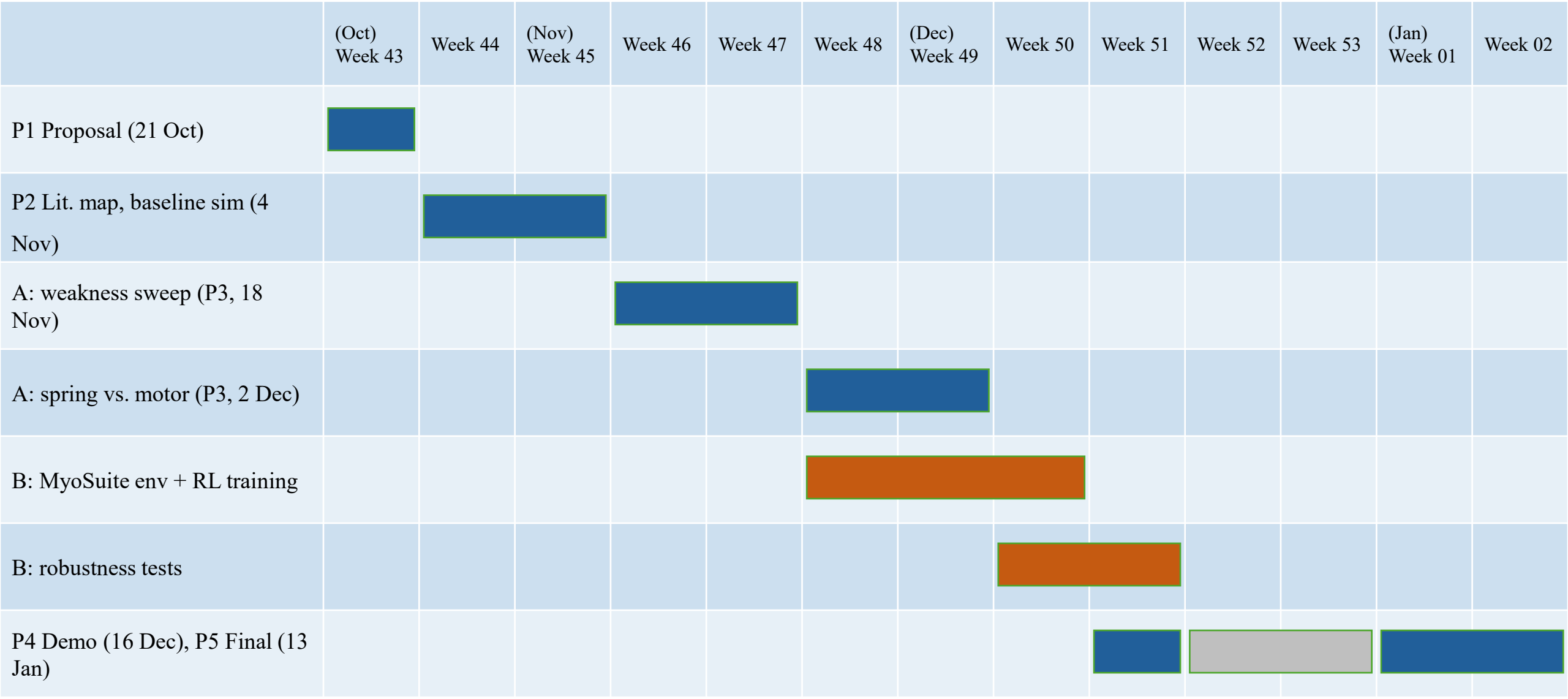
- Model a real wrist exoskeleton (mass, friction, strap fit) instead of an ideal actuator
- Add spasticity and forearm rotation (pronation/supination)
- Detect user intent with EMG/IMU sensor fusion to trigger assistance
- Transfer the learned controller to a wrist exoskeleton test bench (sim-to-real)

Bibliography

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Milestones:

Blue: shared / Route A **Orange:** Route B **Grey:** holiday buffer



Thank You

Repository:
github.com/Rationaliser/wrist-exo-assist-sim